MILITARY SPECIFICATION

TRANSMITTER, RATE GYROSCOPE T-970/AJB-7

1. SCOPE

1.1 This specification covers one type of remote rate-of-turn gyroscope transmitter, designated T-970/AJB-7.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-M-26512	Maintainability Program Requirements for Aerospace Systems and Equipment
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for
MIL-R-27542	Reliability Program for Systems, Subsystems and Equipment
MIL-I-27619	Indicator, Attitude ARU-11/A
MIL-C-38295	Computer Set, Attitude-Reference Bombing AN/AJB-7
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

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MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

3.3 MATERIALS

3.3.1 NONMAGNETIC MATERIALS.- Nonmagnetic materials shall be used for all parts of the transmitter except when magnetic materials are essential.

3.3.2 PROTECTIVE TREATMENT.- When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided. Components contained within hermetic seals shall be considered to be adequately protected against deterioration.

3.3.3 FUNGUS-PROOF MATERIALS.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used, and not within the hermetical seal, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.4 METALS.- Metals used shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuel and salt spray, or atmospheric conditions likely to be met during storage or normal service.

3.3.5 DISSIMILAR METALS.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

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3.3.6 CORROSIVE FUMES.- The materials as installed in the transmitter and under the service conditions specified herein shall not liberate deleterious fumes.

3.4 DESIGN AND CONSTRUCTION

3.4.1 GENERAL DESIGN.- The transmitter covered by this specification is a component of the AN/AJB-7 computer set conforming to MIL-C-38295. It shall be designed to sense the rate of turn of an aircraft by an output signal proportional to the rate at which the aircraft is turning, and shall feed signals to the rate-of-turn needle in the ARU-11/A attitude indicator conforming to MIL-I-27619. The transmitter shall meet all performance requirements specified herein when mounted upside down.

3.4.2 RELIABILITY.- The transmitter shall be so designed that when tested as specified in 4.6.26, the mean-time-between-failures (MTBF) will be at least 5,000 hours. The test program shall be based upon a discrimination ratio of 0.33 and a confidence level of 90 percent. MIL-R-27542 shall be used as a guide for proper implementation of the required reliability into the design and production of the equipment.

3.4.2.1 LONGEVITY.- The transmitter shall have an operating life span (equipment longevity as defined in MIL-T-26667) of not less than 2,000 hours before the equipment consistently fails to meet the MTBF index.

3.4.3 MAINTAINABILITY.- The following maintainability requirements shall be considered in the design of the transmitter.

a. Minimization of the complexity of maintenance tasks (for example calibration, adjustments, inspections, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective part, assembly, or component

d. Minimum number and types of tools and test equipment (special and standard) needed to perform maintenance

e. Optimum accessibility in all systems, equipments, and components requiring maintenance, inspections, removal, or replacement.

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3.4.4 COMPATIBILITY.- The transmitter shall meet all applicable requirements of MIL-C-38295 to insure compatibility within the system.

3.4.5 The transmitter shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

3.4.6 The transmitter shall be so constructed that adjustments and repairs can be made easily by the personnel of overhaul bases.

3.5 PERFORMANCE.- The transmitter shall be capable of satisfactory operation when subjected to any combination of the following conditions:

a. Temperatures - ambient operating temperatures ranging from -62° to +98° C

b. Relative humidity - up to 100 percent

c. Acceleration - acceleration forces up to 20g

d. Altitude - pressures ranging from 30 inches Hg (sea level) to 0.082 inch Hg (approximately 80,000 feet)

e. Fungus - exposure to fungus enzymes

f. Salt fog - exposure to salt fog

g. Sand and dust - exposure to blowing fine sand and dust particles

h. Temperature shock caused by abrupt temperature drop from +85° to -40° C

i. Dynamic shocks at accelerations up to 15g for approximately 11 milliseconds

j. Exposure to acoustic sound fields that are characteristic of aircraft.

3.5.1 RADIO NOISE INTERFERENCE.- The transmitter shall meet the radio noise interference requirements of MIL-I-6181.

3.5.2 WARMUP.- The transmitter shall not require a warmup period of more than 90 seconds under any temperature conditions of the environmental envelope specified herein.

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3.5.3 DIELECTRIC STRENGTH.- The complete electrical system shall be so insulated from the transmitter case that it will withstand 500V dc for a period of 10 seconds.

3.5.4 ELECTRICAL SIGNALS.- The transmitter shall provide electrical signals directly proportional to the rate of yaw of an aircraft.

3.5.5 MAXIMUM RATE OF TURN.- The maximum rate of turn shall be 360° per minute. Higher rates of turn shall cause the sensor to contact the stops.

3.5.6 NATURAL FREQUENCY.- The undamped natural resonant frequency of the rate sensor shall be more than 5 cps.

3.5.7 DAMPING.- The rate sensor shall have a damping ratio between 8 and 12 times the critical ratio.

3.5.8 RESOLUTION AND THRESHOLD.- Both the resolution and the threshold of the sensor shall be less than 0.015° per second.

3.5.9 TRANSIENT RESPONSE.- The response of the transmitter to a step input shall exhibit a time constant of 1.75 ±0.25 seconds.

3.5.10 OUTPUT SIGNAL.- The current output into a 1,000-ohm meter movement shall be 1.0 milliampere (ma) ±6-1/2 percent when the transmitter is rotated in yaw at the rate of 360° per minute, and shall be 0.5 ma ±8 percent when the transmitter is rotated in yaw at the rate of 180° per minute. When roated at 90° per minute, the output shall be 0.250 ma ±9-1/2 percent. When rotated at 45° per minute, the output shall be 0.125 ma ±12-1/2 percent. When the transmitter is rotated to the right in the same manner that an aircraft with the transmitter properly installed would be turned to the right, the polarity of pin C shall be positive. With the precession axis of the transmitter inclined up to 15°, forward end of the transmitter elevated, and the transmitter then rotated in yaw in a horizontal plane, the output signals from the rate transmitter shall be within 3-1/2 percent of the value obtained when the precession axis of the rate transmitter is parallel to the horizontal plane and the transmitter is rotated in yaw in the horizontal plane.

3.5.11 CROSS COUPLING.- When rotated in the pitch axis at maximum rate, the rate transmitter shall show less than 1 percent of the value produced when rotated at the same rate in the yaw axis.

3.5.12 FRICTION AND HYSTERESIS.- The friction and hysteresis in the yaw axis shall not exceed 1 percent of the maximum sensitive rate.

3.5.13 SIGNAL SOURCE IMPEDANCE.- The signal source impedance shall be approximately 1,000 ohms.

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3.6 ELECTRICAL REQUIREMENTS

3.6.1 POWER.- The transmitter shall be designed to operate from 115V single-phase 400-cps ac, and 26V single phase, 400 cps ac power sources as defined in MIL-STD-704. The transmitter shall function satisfactorily with variations in voltage and frequency from 103 to 127V, 23 to 29V, and 380 to 420 cps respectively.

3.6.2 POWER CONSUMPTION.- The peak power consumption of the transmitter shall not exceed 12 va. After a 90-second warmup period, the power consumption shall not exceed 8 va.

3.6.2.1 The power consumption of the pick-off and demodulator circuit shall not exceed 4 va.

3.6.3 RECEPTACLES.- The receptacles provided shall be Bendix PT-02H-10-6P, or equal.

3.6.3.1 PIN UTILIZATION.- The pins of the electrical receptacle shall be wired in the following manner:

Pin	Use
A	Power ground
B	115V single-phase, 400 cps, ac power
C	(+) Output signal (transmitter in right turn)
D	(-) Output signal (transmitter in right turn)
E	Power ground
F	26V, single-phase, 400-cps, ac power

3.6.4 SPIN MOTOR.- The spin motor shall be an ac motor which shall not include commutator, slip rings, contacts, or any similar element that may fail due to wear incurred in normal operation.

3.6.5 UNDERVOLTAGE.- The transmitter shall not be damped by voltage below the minimum specified, and shall automatically resume normal operation when the voltage returns within limits.

3.7 CASE.- The case of the transmitter shall be made of suitable material, shall be uniform in texture, and shall be entirely covered with a durable, dull black finish. Suitable markings shall be provided on the transmitter case to indicate the location of the precession axis, to indicate the forward direction for proper installation in an aircraft, and to indicate the direction of the spin of the gyroscope.

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3.7.1 SEALING PROVISIONS.- The case shall provide a hermetically sealed inclosure for all mechanical and electrical components of the transmitter. The design of the case shall be such that the internal mechanism may be removed from the case as a unit, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the transmitter may be subjected.

3.7.2 FILLING MEDIUM.- Where a hermetically sealed case is used, the filling medium shall be at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65° C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere. Where practical, the 100 percent helium filling medium shall be utilized. If the gyro is a floated unit and is sealed in a compartment of the case separate from other components, this requirement shall apply only to the compartment containing the other components.

3.8 PART NUMBERING OF INTERCHANGEABLE PARTS.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.9 MOUNTING SCREWS.- The contractor shall furnish sufficient 8-32NC2-A mounting screws for installing the transmitter. They shall have a durable, dull black, oxidized or black nickel finish. The length shall be sufficient for mounting on a base 3/8-inch thick.

3.9.1 SCREW THREADS.- Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.9.2 ENVELOPE.- An envelope containing mounting screws shall be furnished by the contractor and shall be packed with each instrument. The envelope shall be marked as follows:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

3.10 SHOCK MOUNT.- A shock mount shall be provided between the assembly and the airframe.

3.11 STOPS.- Suitable stops shall be mounted in the transmitter to prevent damage in extreme positions.

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3.12 DIMENSIONS.- The dimensions of the transmitter shall be in accordance with figure 1.

3.13 WEIGHT.- The total weight of the transmitter, completely assembled and less mounting screws and cables, shall not exceed 3 pounds.

3.14 FINISHES AND PROTECTIVE COATINGS

3.14.1 ALUMINUM ALLOY PARTS.- Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.14.1.1 Small holes and case inserts need not be anodized.

3.14.1.2 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.14.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 shall be used in lieu of anodizing.

3.14.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.14.1.5 Whem abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

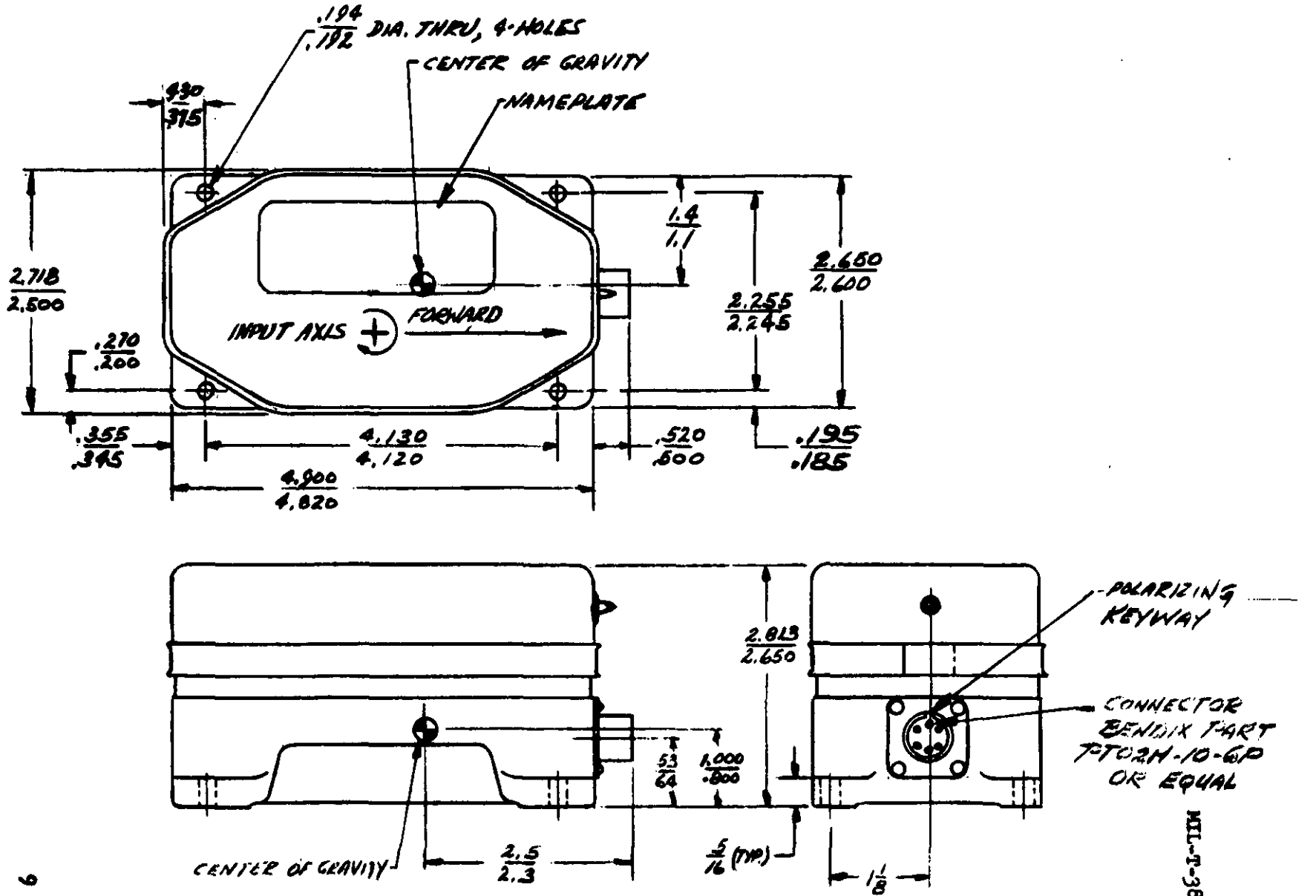
3.14.2 CADMIUM PLATING.- Cadmium plating shall be in accordance with QQ-P-416 and of a class that is adequate to achieve the degree of protection required.

3.15 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The direction of spin of the gyro rotor shall be clearly marked on the case.

3.16 WORKMANSHIP.- The transmitter, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs, sharp edges, and resin flash that may crumble.

3.16.1 DIMENSIONS AND TOLERANCES.- Dimensions and tolerances not specified shall be as close as is consist with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the transmitter, they shall be held or limited accordingly.

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DIMENSIONS IN INCHES
TOLERANCE:- FRACTIONS ± 1/32

FIGURE 1 TRANSMITTER

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3.16.2 SCREW ASSEMBLIES.- Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.16.3 CLEANING.- The transmitter shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF TESTS.- The inspection and testing of transmitters shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 TEST CONDITIONS

4.3.1 STANDARD ATMOSPHERIC CONDITIONS.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test will be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25° C). When tests are made with the atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 VOLTAGE INPUT.- Unless otherwise specified, the transmitter shall be tested with the input specified in 3.6.3.1.

4.3.3 ATTITUDE.- Unless otherwise specified, the transmitter shall be tested in its normal operating position.

4.3.4 METER MOVEMENT.- Unless otherwise specified, the transmitter shall be tested while connected to a 1,000-ohm load simulating the meter movement specified in 3.5.10.

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4.3.5 TEST SEQUENCE.- Unless otherwise specified, tests shall be conducted in the sequence listed under 4.6.

4.3.6 FRICTION REMOVER.- An Edwards vibrator, or equivalent, may be used to remove friction before a test reading is taken. Tapping the instrument will not be acceptable. Friction within the instrument shall be sufficiently low that it can be removed by the vibrator. If the vibrator is used for any one test, it shall be used throughout the testing specified under 4.6.

4.4 PREPRODUCTION TESTING

4.4.1 TEST SAMPLES.- The test samples shall consist of 14 transmitters representative of the production equipment. Three of the transmitters shall be subjected to all the tests except reliability and the other 11 transmitters shall be subjected to the reliability test specified in 4.6.26.

4.4.1.1 RELIABILITY DEMONSTRATION.- Unless otherwise specified, 11 transmitters shall be selected at random from the first 50 produced and tested as specified in 4.6.26 for a total of 16,500 hours, unless rejection is determined in accordance with figure 2 prior to that time.

4.4.2 TEST REPORT.- When the tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity:

a. Test report - Three copies of a test report in accordance with MIL-STD-831

b. The samples which were tested

c. Two sets of assembly drawings

d. Brief operating instructions.

4.4.3 PREPRODUCTION TESTS.- The preproduction tests shall consist of all tests described under 4.6.

4.5 ACCEPTANCE TESTS.- Acceptance tests shall consist of the following:

a. Individual tests

b. Sampling plans and tests.

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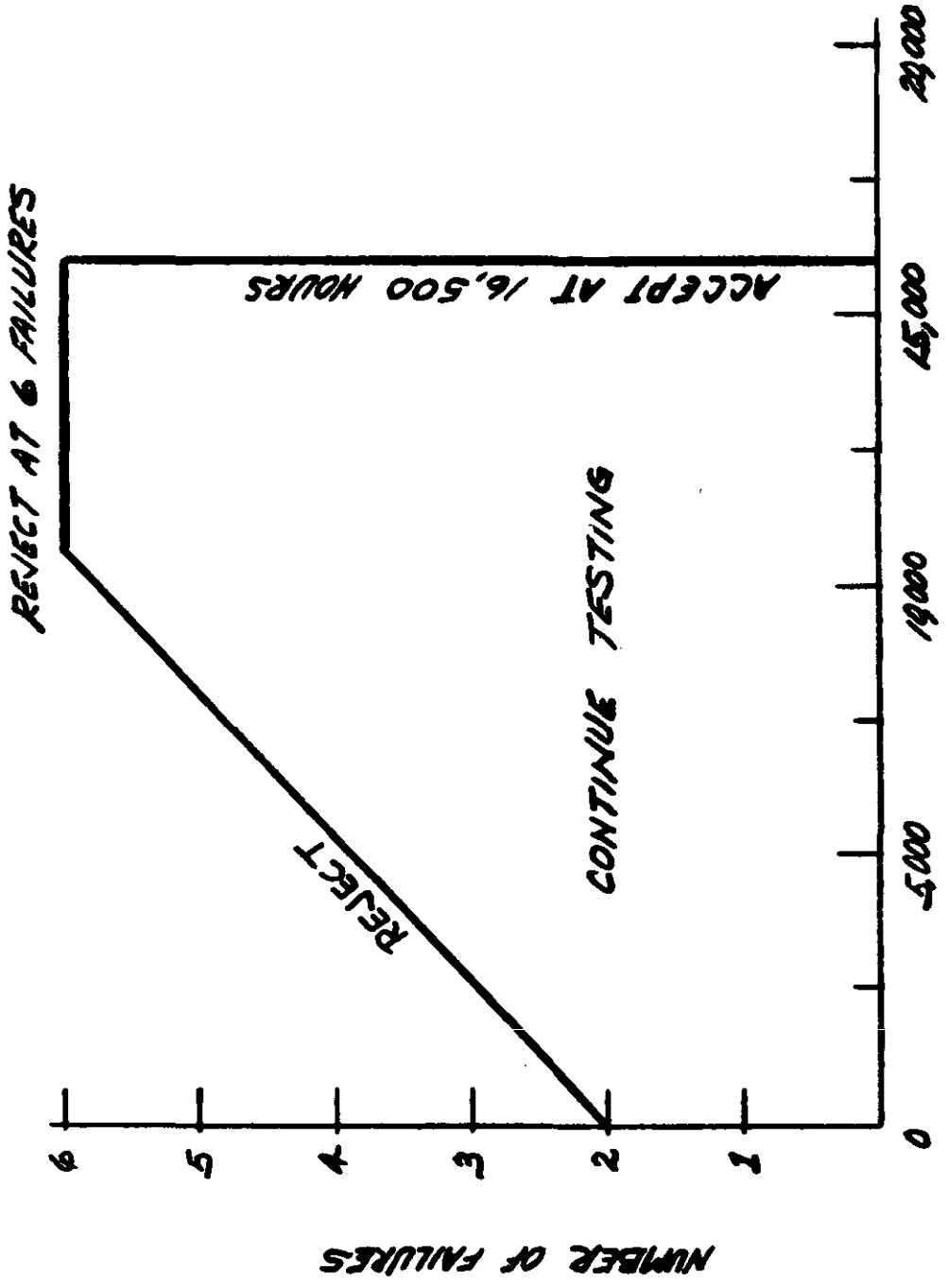


FIGURE 2 CONTINUE TEST - REJECT CRITERIA

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4.5.1 INDIVIDUAL TESTS.- Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Power consumption

c. Null limits

d. Sensitivity

e. Transient response

f. Cross coupling

g. Sealing

h. Dielectric

i. Voltage extremes

j. Resolution and threshold

k. Friction and hysteresis

l. Damping factor

4.5.2 SAMPLING PLANS AND TESTS

4.5.2.1 SAMPLING PLAN A.- One transmitter shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Sensitivity at low temperature

c. Sensitivity at high temperature

d. Vibration error

e. Yaw oscillation.

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4.5.2.2 SAMPLING PLAN B. - Unless otherwise specified, 3 transmitters shall be selected at random from the first 15 items on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Radio noise interference

c. High temperature

d. High altitude - low temperature

e. Temperature shock

f. Dynamic shock

g. Vibration failure

h. Acceleration

i. Humidity

j. Fungus

k. Salt fog

l. Sand and dust

m. Sealing thermal shock.

4.5.2.2.1 EMPTY CASES. - At the option of the procuring activity, the humidity, salt fog, sand and dust, fungus, and sealing thermal shock tests may be conducted on empty cases with all external parts attached as on production units.

4.5.2.3 REJECTION AND RETEST. - When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 INDIVIDUAL TESTS MAY CONTINUE. - For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification.

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4.5.3 SAMPLING PLAN C.- Unless otherwise specified, 9 transmitters shall be selected at random from each 50 or less produced on the contract or order and tested as specified in 4.6.26 for a total of 5,500 hours.

4.5.3.1 ACCEPTANCE OR REJECTION.- Figure 3 shall be the criteria used to determine whether the equipment will be accepted or rejected. Certified time records shall be maintained of the test time accumulated on the equipment after starting the environmental cycling specified in 4.6.26. In the event a continue-test decision applies after 5,500 hours of testing, the production quantity shall be considered as conditionally acceptable; however, the accumulated time and failures shall be added to the data of the next production quantity for a composite accept-reject decision. If the continue-test condition persists, the data from successive production quantities shall be accumulated until 6 failures or 16,500 hours of testing have occurred at which time an accept-reject decision shall be made.

4.5.3.2 RECORDING.- Recording, data handling, and reporting procedures shall be in accordance with MIL-R-26667.

4.6 TEST METHODS

4.6.1 EXAMINATION OF PRODUCT.- The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, and configuration as shown on figure 1.

4.6.2 POWER CONSUMPTION.- The transmitter shall be connected for normal operation, without power applied and with the gyro rotor at rest. Power shall be applied and the starting power, measured within 5 seconds from the time power is applied, shall not exceed 12 va to the gyro rotor. Ninety seconds subsequent to application of power, the motor power consumption shall not exceed 8 va. At both the 5-second and 90-second conditions, power consumption of the pick-off and demodulator circuit shall not exceed 4 va.

4.6.3 NULL LIMITS.- When the output signal is fed through a 1,000-ohm load, the output current with 0° per minute rate of turn shall be zero within ±7.5 microamperes (ua).

4.6.4 SENSITIVITY.- The output of the rate sensor, in both the clockwise(cw) and counterclockwise (ccw) directions, shall be as follows for inputs about the yaw axis. These readings shall be taken with the output signal fed into a 1,000-ohm, ±3 percent, resistance.

Rate of Turn (Degrees per minute)	Output (ma)
45	0.125 ±12-1/2 percent
90	0.250 ±9-1/2 percent
180	0.50 ±8 percent
360	1 ±6-1/2 percent

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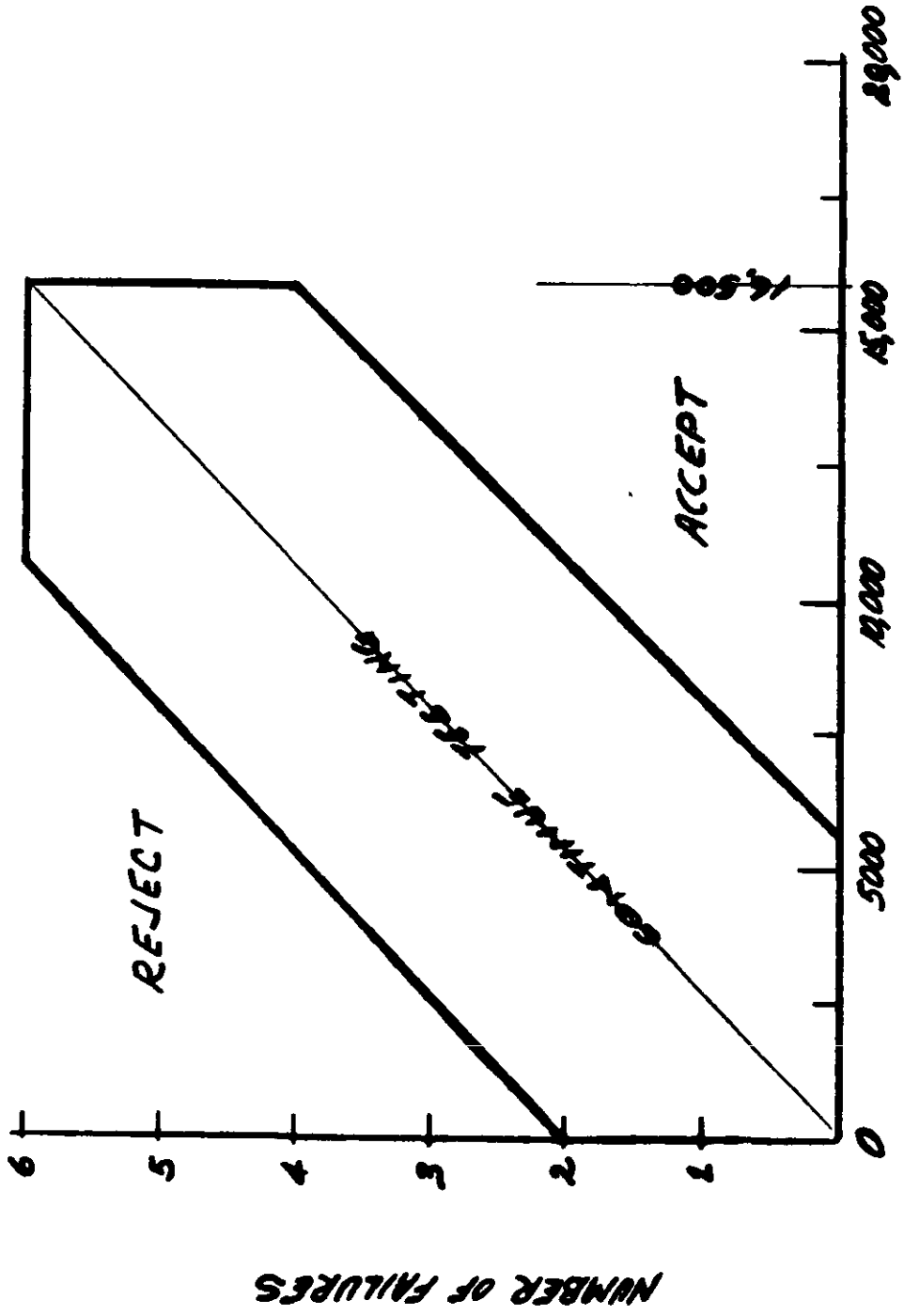


FIGURE 3 - ACCEPT - CONTINUE TEST - REJECT CRITERIA

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4.6.5 TRANSIENT RESPONSE.- While the rate transmitter is rotating at 360° per minute in either direction and the output is as specified in 4.6.4, the turning shall be stopped suddenly. The time for the output current to drop to 1.47 ma shall be measured and shall be within 1.75 ±0.25 seconds. The test shall be repeated in the opposite direction of rotation.

4.6.6 CROSS COUPLING.- The rate transmitter shall be rotated in each direction about the pitch axis at a rate of 360° per minute. The output reading shall not exceed 10 ua.

4.6.7 SEAL.- The transmitter shall be tested for leaks by means of a mass-spectrometer type of helium leak detector. The differential pressure existing when the test is conducted shall be the same as that to which the instrument is filled, approximately 1 atmosphere. The leak rate shall be based on a 1-atmosphere differential pressure with pure helium as the filling medium. Under these conditions (1-atmosphere differential pressure and pure helium as the filling medium), the leak rate shall not exceed 0.10 micron cubic foot per hour. Equivalent helium leak rates for the same hole size at other differential pressures and helium-nitrogen mixtures, or both, shall be calculated by the manufacturer. The leak test shall be conducted either with the instrument evacuated and surrounded by helium, or after the instrument has been filled. If the gyro is a floated unit, this test shall not apply to that compartment.

4.6.8 DIELECTRIC.- The completely assembled transmitter, with filter installed, shall be subjected to a potential of 500V dc between the isolated terminals for a period of 10 seconds. There shall be no breakdown of insulation or any permanent damage to the instrument.

4.6.9 VOLTAGE EXTREMES.- Voltages of 103 and 127V ac shall be applied to the transmitter at 380 and 420 cps. While each voltage is applied, the test specified in 4.6.4 shall be repeated except that all readings shall be within +10, -17 percent of the normal output values at the respective rate of turn.

4.6.10 RESOLUTION AND THRESHOLD.- Both the resolution and threshold shall be tested with a rate of 0.015° per second or more so that the perceptable change in output signal will be at least 1.5 ua.

4.6.11 FRICTION AND HYSTERESIS.- The friction and hysteresis shall be tested and the total variation at zero input resulting from subjecting the unit to the maximum rate of turn as defined in 3.5.5, in both cw and ccw directions, shall not exceed 10 ua.

4.6.12 DAMPING FACTOR.- The damping factor of the sensor shall be determined and shall be within the limits of 8 to 12 times the critical ratio.

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4.6.13 SENSITIVITY AT LOW TEMPERATURE.- The transmitter shall be properly connected and placed in a chamber at a temperature of -54° ±2° C for a period of 4 hours. At the end of the 4 hours and while still at the low temperature, power shall be applied to the transmitter. After power has been applied for 10 minutes, the transmitter shall meet the test specified in 4.6.4, except the tolerance shall be increased by 3-1/2 percent. When at rest and at low temperature subsequent to rotation, the transmitter shall meet the test specified in 4.6.3, except the tolerance shall be zero ±14 ua.

4.6.14 SENSITIVITY AT HIGH TEMPERATURE.- The transmitter shall be properly connected and subjected to a temperature of 71° ±2° C for a period of 4 hours. While still at this temperature, power shall be applied. After 5 minutes of operation, the transmitter shall be subjected to and shall pass the test specified in 4.6.4, except the tolerance shall be increased by 3-1/2 percent. While still at the high temperature, the transmitter shall pass the test specified in 4.6.3, except the tolerance shall be ±14 ua. The temperature shall be increased to 98° C and held for 10 minutes. During this time, the transmitter shall be running and shall indicate the imposed rate of turn within the limits specified.

4.6.15 VIBRATION ERROR.- The vibration error test shall be conducted with power applied. It shall consist of a frequency survey with vibration applied to the longitudinal, lateral, and vertical axes of the transmitter. The transmitter shall be subjected to vibration with a constant applied double amplitude of 0.060 inch through the frequency range of 10 to 55 cps in each of the 3 mutually perpendicular axes. The duration of the vibration shall be 5 minutes in each axis for one survey of the complete frequency range or 5 minutes with vibration applied in a circular motion for one survey of the complete frequency range. During this test, the output signal shall not exceed 50 ua when fed into a 1,000-ohm resistance.

4.6.16 YAW OSCILLATION.- The transmitter, with power applied for not less than 90 seconds, shall be subjected to the following yaw oscillations. The average value of the signal integrated over at least 10 full cycles shall not exceed 65 ua.

a. 2° double amplitude for an 8-second period

b. 4° double amplitude for an 8-second period.

4.6.16.1 STATIC BALANCE.- The transmitter shall be mounted upside down in the test fixture and subjected to the tests specified in 4.6.3 through 4.6.6.

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4.6.17 RADIO NOISE INTERFERENCE.- The transmitter shall pass the radio noise interference test, for devices of this type, specified in MIL-I-6181.

4.6.18 HIGH TEMPERATURE.- The high temperature test shall be conducted in accordance with method 501 of MIL-STD-810 for a period of 24 hours. While still at the high temperature, rated power shall be applied. The transmitter shall operate without sticking. After the transmitter has returned to room temperature, it shall pass the test specified in 4.6.4. There shall be no damage that would affect subsequent operation.

4.6.19 TEMPERATURE-ALTITUDE.- The transmitter shall be subjected to a temperature-altitude test in accordance with MIL-STD-810, method 504, class 2, except the altitude shall be 80,000 feet, in step 5 the stabilized temperature shall be maintained for 4 hours, and in step 7 the temperature shall be 85°. The transmitter shall operate without sticking. After returning to room temperature and atmospheric pressure, the transmitter shall meet the individual tests. The transmitter shall operate properly throughout this test and there shall be no damage that would affect subsequent operation.

4.6.20 TEMPERATURE SHOCK.- The temperature shock test shall be conducted in accordance with method 503 of MIL-STD-810. After the transmitter has returned to room temperature, the connector and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals. The transmitter shall then be subjected to and shall pass the individual tests.

4.6.21 DYNAMIC SHOCK.- The dynamic shock test shall be conducted in accordance with method 516, procedures I and IV of MIL-STD-810. Following this test, the transmitter shall be subjected to and shall meet the individual tests.

4.6.22 VIBRATION FAILURE.- The vibration failure test shall be conducted in accordance with method 514, curve B, of MIL-STD-810, with power applied. At the end of this test, the transmitter shall be subjected to and shall pass the test specified in 4.6.4. No damage or irregular operation shall result from this test.

4.6.23 ACCELERATION.- The transmitter, not operating, shall be mounted on a centrifuge in its normal operating position and accelerated, first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. The transmitter shall be subjected to an acceleration of 20g in each of these axes for a period of 1 minute. At the end of the acceleration, the transmitter shall be subjected to the individual tests. No damage to the transmitter shall result from this test.

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4.6.24 ENVIRONMENTAL TESTS.- The transmitter shall be subjected to the following environmental tests in accordance with the specified procedures of MIL-STD-810. At the option of the procuring activity, these tests may be performed on finished empty cases. At the completion of each test, the transmitter case shall be inspected for damage that would impair transmitter operation.

4.6.24.1 HUMIDITY.- The humidity test shall be conducted in accordance with method 507, procedure I.

4.6.24.2 FUNGUS.- The fungus test shall be conducted in accordance with method 508, procedure I.

4.6.24.3 SALT FOG.- The salt fog test shall be conducted in accordance with method 509, procedure I, except that the test period shall be not less than 168 hours.

4.6.24.4 SAND AND DUST.- The sand and dust test shall be conducted in accordance with method 510, procedure I.

4.6.25 SEALING THERMAL SHOCK.- The transmitter case, with mechanisms removed and sealed in the same manner as the complete transmitter, shall be immersed alternately in tap water maintained at 85° ±5° C and at 5° ±4° C for a total of 8 cycles. The length of time for each immersion shall be 30 minutes with not more than 5 seconds elapsing between immersions. No damage to the hermetic seal shall occur as a result of this test. Following the immersions, the transmitter case shall be subjected to and shall meet the test specified in 4.6.7.

4.6.26 RELIABILITY.- The transmitter shall be properly connected with power supplied as specified in 3.6.3.1 and subjected to the reliability environment cycles specified on figure 4. After each 160 hours of operation, the equipment shall meet the tests specified in 4.6.3 through 4.6.6.

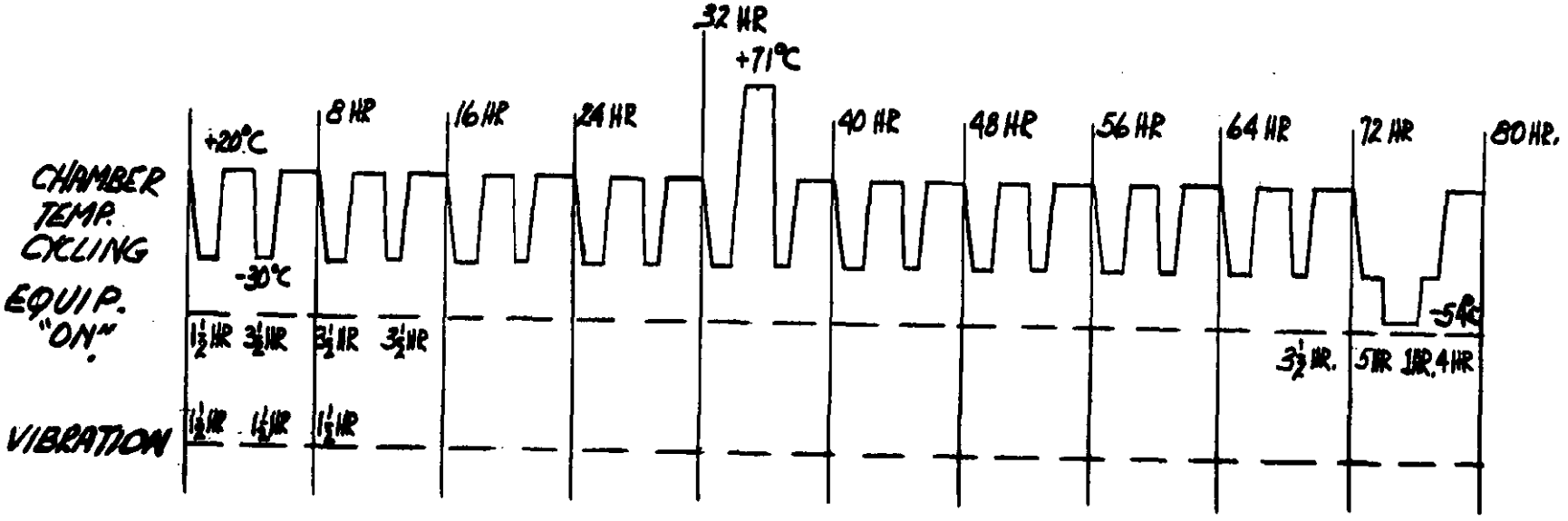
4.6.26.1 LONGEVITY.- At the completion of the test specified in 4.6.26, three of the transmitters shall continue to be tested for a total of 2,000 hours each.

4.6.26.2 RELEVANCY CRITERIA.- All failures shall be classified as relevant unless approved for exclusion by the procuring activity.

4.6.26.3 TEST INFORMATION.- Sixty days prior to starting this test, the contractor shall submit detailed test information to the procuring activity for approval. This information shall include but not necessarily be limited to:

a. Equipment used in monitoring performance

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NOTES:

a. Environmental cycle repeats every 80 hours.
b. Temp rate of change of chamber shall be 0.1°C - 0.5°C per second.
c. Vibration level 0.2G at 20 to 30 cps.
d. "Off" periods for equipment "On" and "Off" are ½ hour (except 76th hour).
e. Monitoring of system outputs for catastrophic-type failures performed during each "ON" cycle.
f. "Satisfactory performance" tests on systems and individual units conducted every 160 hours of environment cycle to detect performance deviation type failures.

FIGURE 4 RELIABILITY TEST ENVIRONMENT CYCLES

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b. Periodicity of calibration checks to be made on monitoring equipment

c. Description of environmental test facilities to be used, including recording facilities, method of control of heating and cooling facilities, and vibration equipment

d. Frequency of monitoring.

4.6.26.4 TEST RECORDS.- Throughout this test, the contractor shall maintain adequate test records which shall be prepared in a form suitable for reproduction. As a minimum, the records shall include the following in accordance with MIL-R-26667: An operating sheet, log of failures, failure report, equipment repair sheet, failure tag, and failed-part summary analysis. Reproducible copies of the data shall be submitted to the procuring activity within 30 days after the completion of each test phase.

4.6.26.5 REPAIR OF FAILED UNITS.- In the event of a failure during testing, the failed unit shall be repaired and the test continued.

4.6.27 MAINTAINABILITY VERIFICATION.- Maintainability verification shall be in accordance with MIL-M-26512, appendix A, paragraph entitled "Maintenance Task Test and Demonstration". All tasks selected shall be effectively simulated.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 INTENDED USE.- The T-970/AJB-7 transmitter covered by this specification is intended for use in the F4C aircraft to sense the rate at which the aircraft turns.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted

c. When sampling plan C test will not be conducted

d. Levels of packaging and packing required.

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6.2.1 FAILURE REPORT.- During testing, a report of each failure that occurs should be submitted to the procuring activity within 2 working days after the failure.

6.3 DEFINITION

6.3.1 HERMETIC SEAL.- A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic, or both, materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3.2 FAILURE.- Where a condition is specified as a requirement, a failure is considered as any electrical or mechanical condition that fails to comply with the requirements specified herein.

Custodian:
Air Force - 11

Preparing Activity:
Air Force - 11

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

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